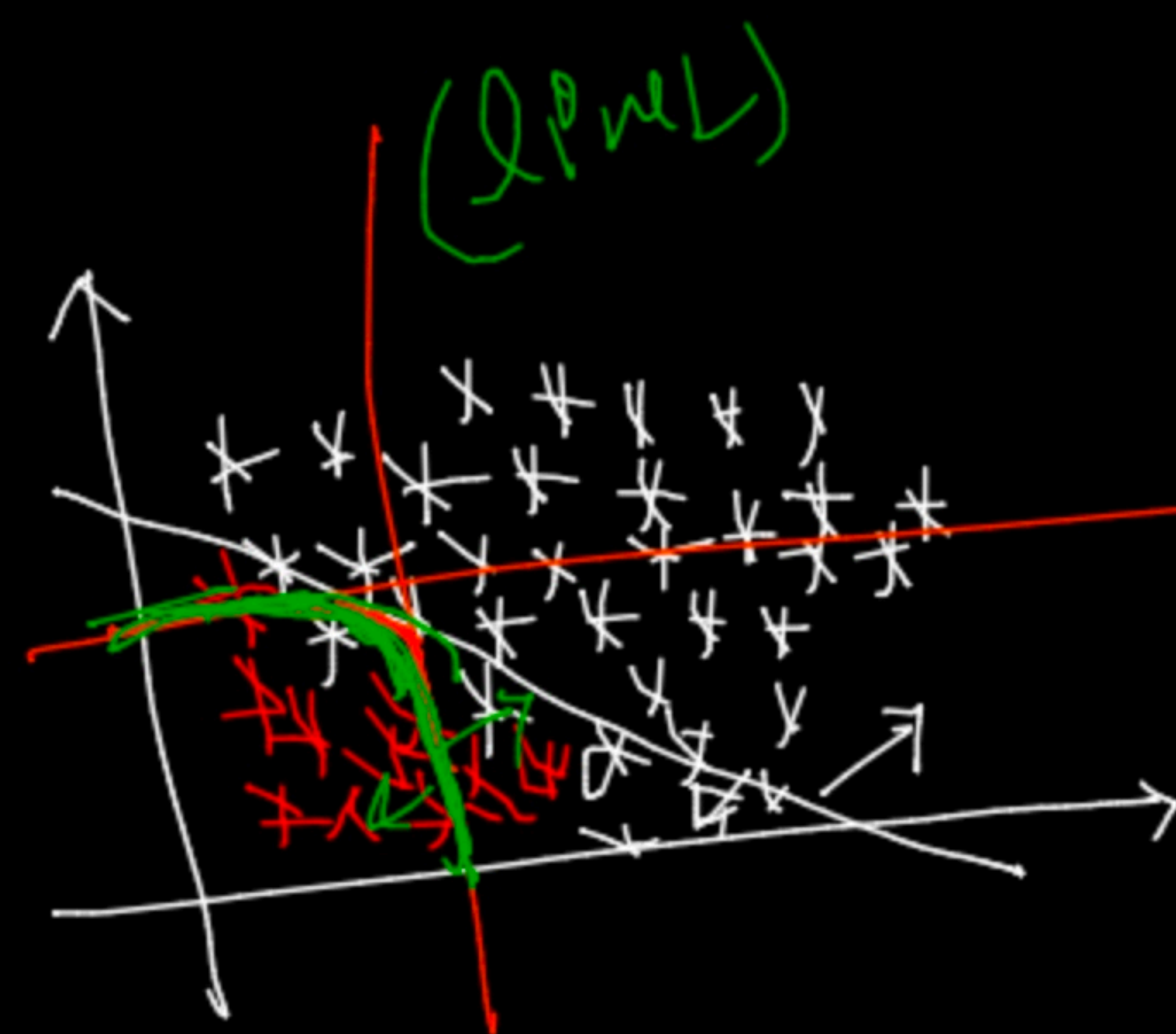




line

non-linear



log P

ML

↓
(NN) sleeples

text $\rightarrow$ vector $\rightarrow$ MLDL

(1) Statistical way

mathematical $\rightarrow$

'self'
TF-IDF

Hello

H₁ how are you
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 w_1 w_2 w_3 w_4

✓	✓	✓	✓	✓
---	---	---	---	---

text $\rightarrow$ vector

$\rightarrow$ 'word embedding'

google

w2vec

(data)

(pre-trained)

text $\rightarrow$ w₁ w₂ w₃ w₄ w₅

ML

$O(N \times T)$

'x'

Internet

↓
wikipedias (topics)

knowing → 784

2014

↑
19ms/m

↑
Internet

↑
play

!!
{ "hello" → EM() →

"hi" → EM

world

small

1
2
3

~784'
K
[0.4 | 0.1 | 2 | - | - | - | -]

[| | | | | | |]

OpenAI (Gen AI)

embedding → (large data)

powerful

cleaning / stop words / etc → vector

text

sentiment

w1vec

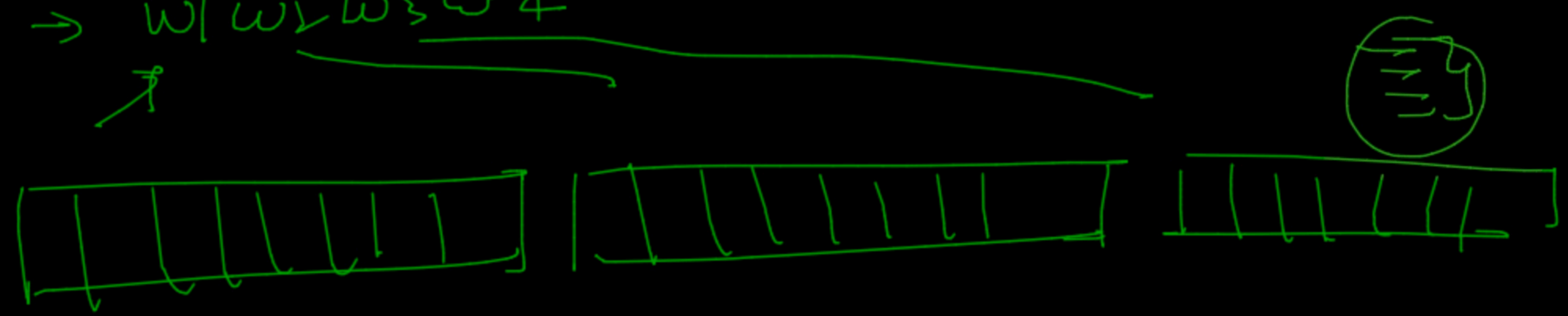
→ tfidf

↳ score

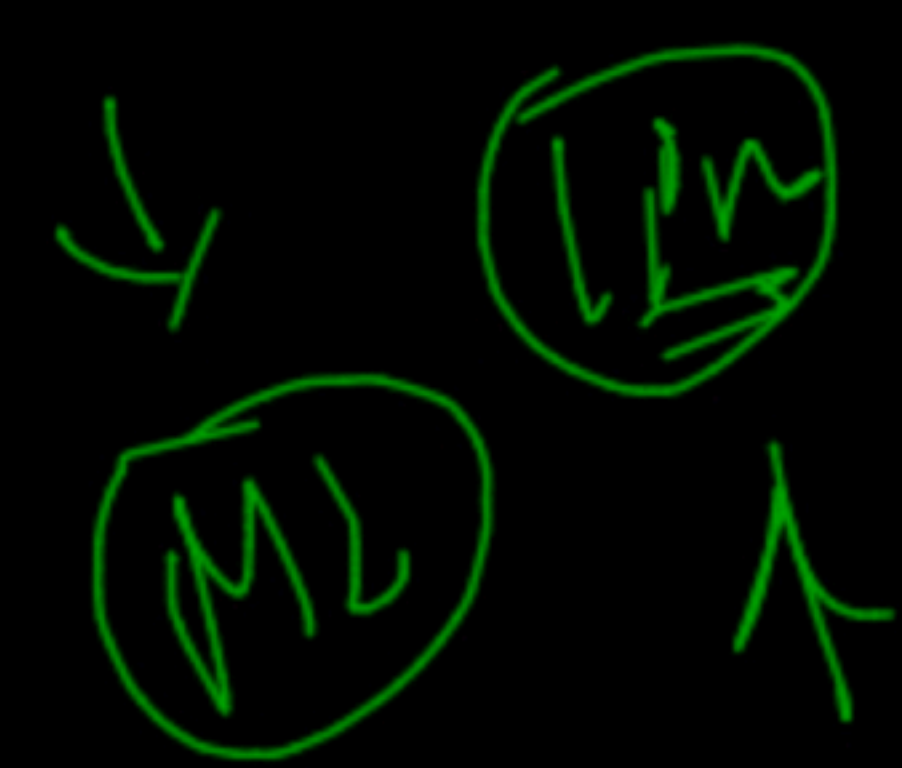
② w2vec

↳

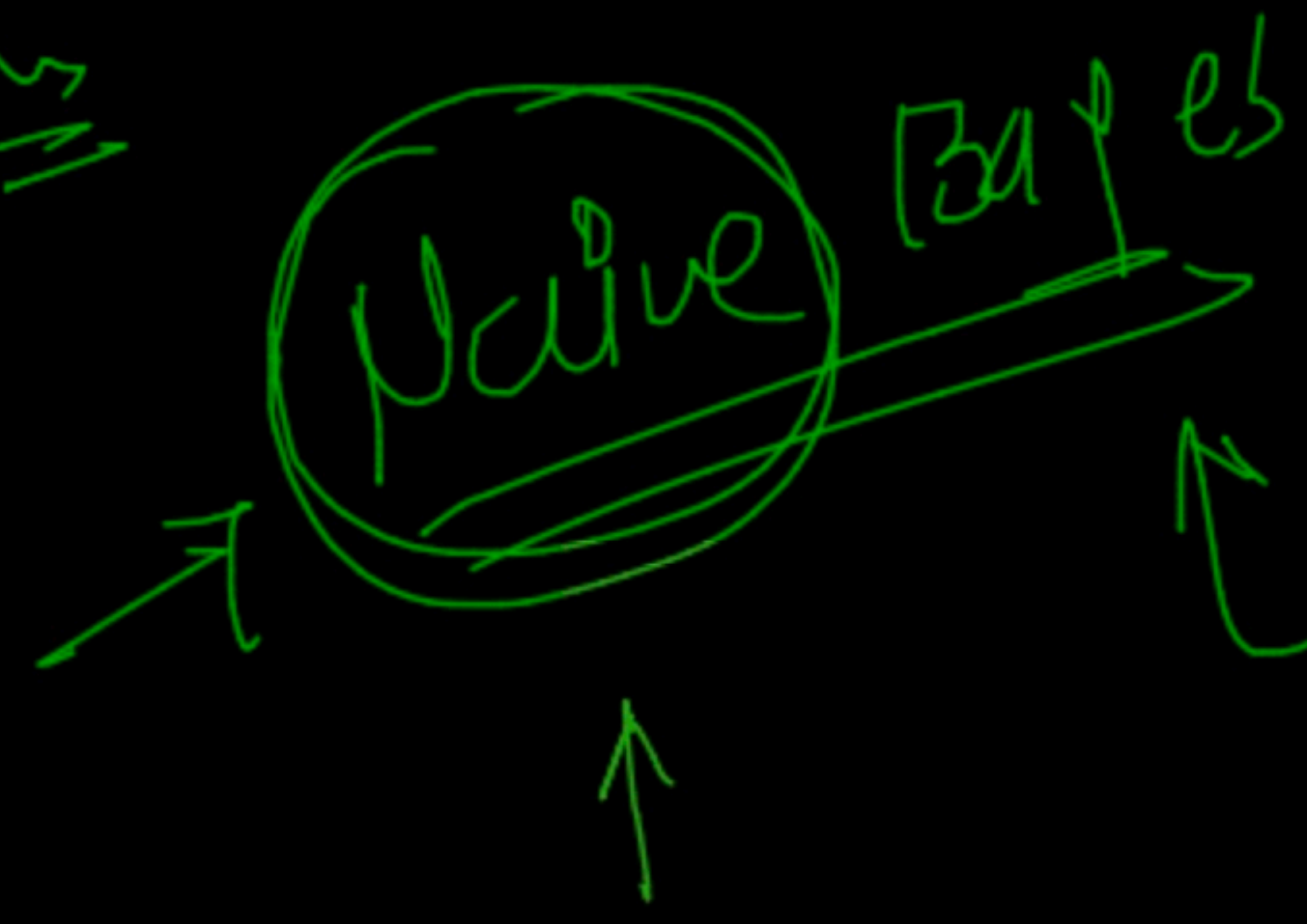
text $\rightarrow w_1 w_2 w_3 w_4$



NLP

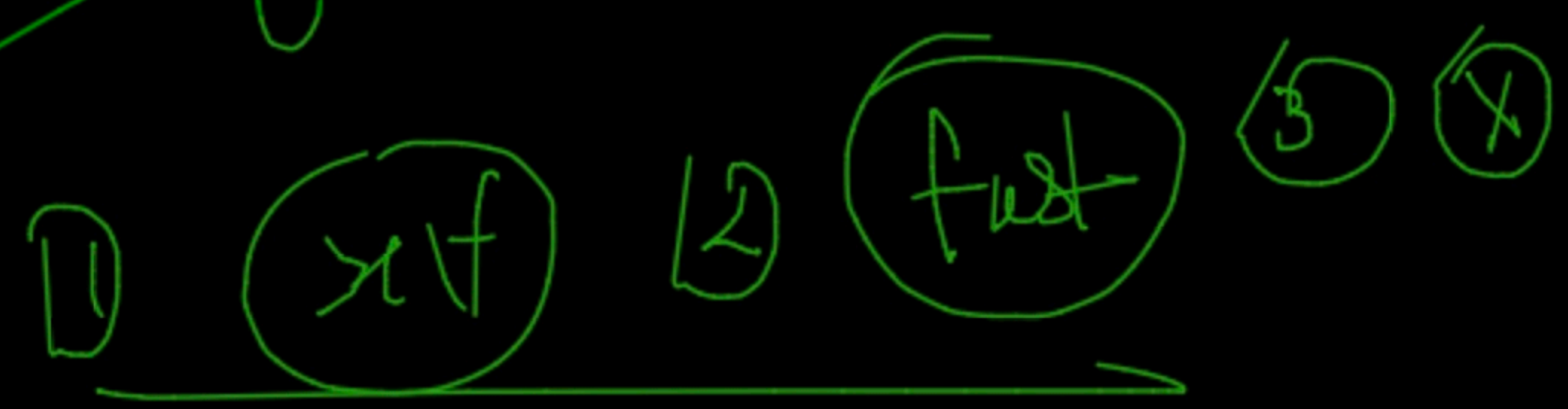


ms



text classification

email
sentiment



Naive Bayes

prob → conditional prob.
↑

$\begin{cases} A \\ B \end{cases} \rightarrow \underline{\text{given}}$
↘ ☒

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

condition

↓

Bayes theorem

posterior

$$\rightarrow P(A|B) = \frac{\overbrace{P(B|A) \cdot P(A)}^{\text{likelihood} \quad \text{prior}}}{\underbrace{P(B)}_{\text{evidence}}}$$

$\begin{cases} \text{pt1} \rightarrow w_1 w_2 w_3 \rightarrow +ve \\ \text{pt2} \rightarrow w_3 w_4 w_1 \rightarrow -ve \\ \text{pt3} \rightarrow w_4 w_1 w_2 \rightarrow +ve \end{cases}$

$$P(w_1 | +ve) = \frac{\# \text{pts when } (w_1) \& y = +ve}{\text{total \# pts}}$$

Training

$$\begin{cases} \underline{P(+ve)} = \frac{2}{3} \Rightarrow 66\% \\ P(-ve) \Rightarrow 0.33 \end{cases}$$

$$\begin{cases} P(w_1 | +ve) = \frac{2}{2} \rightarrow 100 \\ P(w_2 | +ve) = \frac{2}{2} \rightarrow 100 \\ P(w_3 | +ve) = \frac{1}{2} \rightarrow 0.5 \\ P(w_4 | +ve) = \frac{1}{2} \rightarrow 0.5 \end{cases}$$

$$P(Y=1 \mid \underset{\substack{\uparrow \\ \text{email}}}{\text{text}}) = P(Y=1 \mid \underset{\substack{\downarrow \\ \text{words}}}{w_1 w_2 w_3 \dots w_d})$$

$$P(Y=1) * \overbrace{P(w_1 \mid Y=1) * P(w_2 \mid Y=1) * \dots * P(w_d \mid Y=1)}$$

Key idea

$$P(Y=1 \mid \text{text}) = P(Y=1) \prod_{i=1}^d P(w_i \mid Y=1)$$

$$P(Y = \text{negative} | \text{text}) \Rightarrow P(Y = 0 | w_1 w_2 \dots w_d)$$

$\Downarrow$

$$P(Y = 0) \prod_{i=1}^d P(w_i | Y = 0)$$

✓

$\Uparrow$

Prediction
↑

text

↓

$w_3 w_4 w_1$

→

$P(Y=1 | \text{text})$

$w_3 w_4 w_1$

↑

$P(Y=1)$

↑

$$= P(Y=1) * P(w_3 | Y=1) * P(w_4 | Y=1) * P(w_1 | Y=1) \dots$$

$$= 0.66 * 0.5 * 0.5 * 0.12$$

$P(Y=1 | \text{text})$

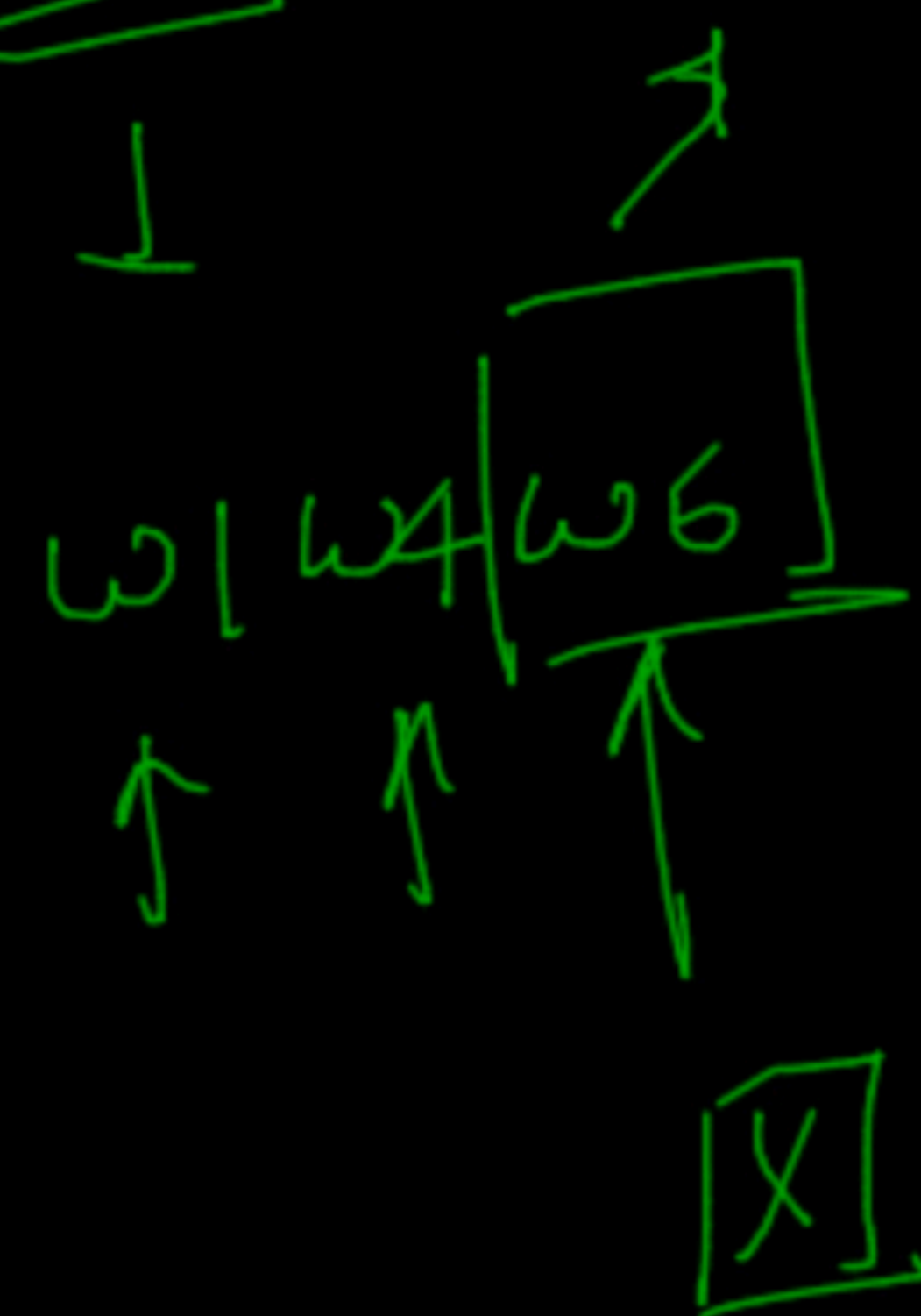
=

8pm'12 →

Prediction

↑

test



$$P(Y=1 | \text{test})$$

⇓

$$P(Y=1) * P(w_1 | Y=1) * P(w_2 | Y=1) * P(w_6 | Y=1)$$

The equation is annotated with red markings: a bracket under $P(Y=1)$ with an arrow pointing to '0', a bracket under $P(w_1 | Y=1)$ with an arrow pointing to '0', a bracket under $P(w_2 | Y=1)$ with an arrow pointing to '0', and a red circle around $P(w_6 | Y=1)$ with an arrow pointing to '0'.

→ { P/Y

'0'

Quick

'Laplace smoothing'

$$\left\{ \begin{array}{l} \frac{0}{2} \rightarrow \\ \frac{0 + \overbrace{0.001}}{2} \end{array} \right. \begin{array}{l} \\ \text{error} \\ \uparrow \end{array}$$

↓

$$P(w' | Y=1) = \frac{\text{\# train pts } w' \text{ \& } Y=1}{\text{\# pts when } Y=1}$$

$$\Rightarrow \frac{0}{2} \Rightarrow \frac{0}{0}$$

$$= \frac{0 + \alpha}{n_1 + kd}$$

$$n_1 + kd$$

$k = \# \text{ distinct value } (w) \text{ can take}$

$$\boxed{\alpha} \rightarrow$$

$$p(w' | y=1) = \frac{0 + \alpha}{\underline{100} + 2\alpha}$$

$$\boxed{\alpha=1}$$

$$\frac{1}{100+2} \neq 0$$

Q2 $\alpha = \underline{\underline{10,000}}$

$\nearrow P(w' | Y=1) = \frac{0 + 10,000}{100 + 20,000} \approx \left(\frac{1}{2}\right) \quad (0.5)$

$\boxed{\alpha = 1}$

high

$\boxed{\alpha = 1} \rightarrow \boxed{\underline{\underline{high}}}$

$$P(Y=1 | w_1 w_2 w_3 \dots w_d)$$

↓

review

no - (50)

skew

$$\log \left(P(Y=1) * P(w_1 | Y=1) * P(w_2 | Y=1) \dots \right)$$

log prob

$$0.1 * 0.02 \dots$$

↓

$$0.00004 \rightarrow$$

↓
10⁻⁵

$$\log(0.0004) \Rightarrow -2.23$$

↑

$$\log \left(\frac{P(Y=1)}{P(Y=-1)} \right) = \frac{d}{1} \log \left(\frac{P(w_i | Y=1)}{P(w_i | Y=-1)} \right) \quad (\text{case i})$$

$$n \geq 2000 \begin{cases} 1000 (+ve) \\ 1000 (-ve) \end{cases}$$

Bias-variance tradeoff

High Bias $\rightarrow$ underfitting

High var $\rightarrow$ overfit

good $\rightarrow$ ~~(w)~~ ✓
good $\rightarrow$ ~~(-w)~~ ✓

α

$$\alpha \geq 0$$

$$P(w_i | Y=1) = \frac{2}{1000}$$

'smallest'

$\downarrow$
overfitting